

Dynamic regression

08

Outline

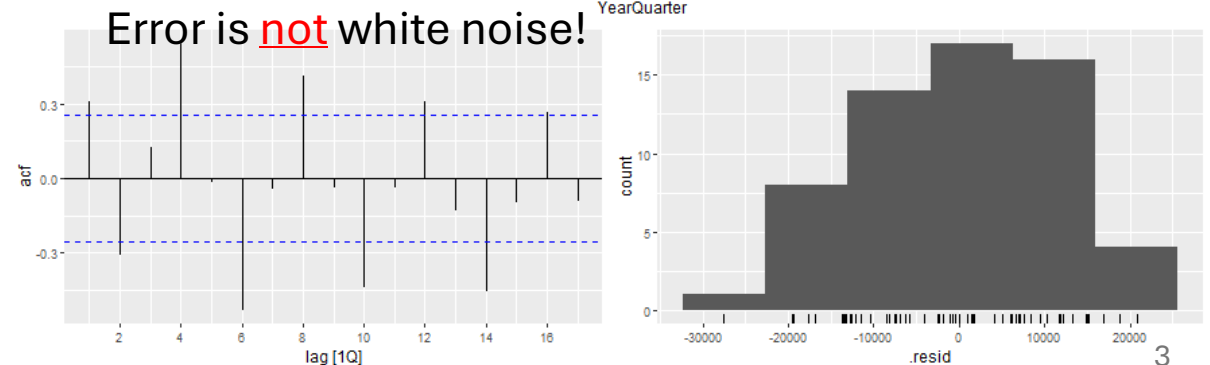
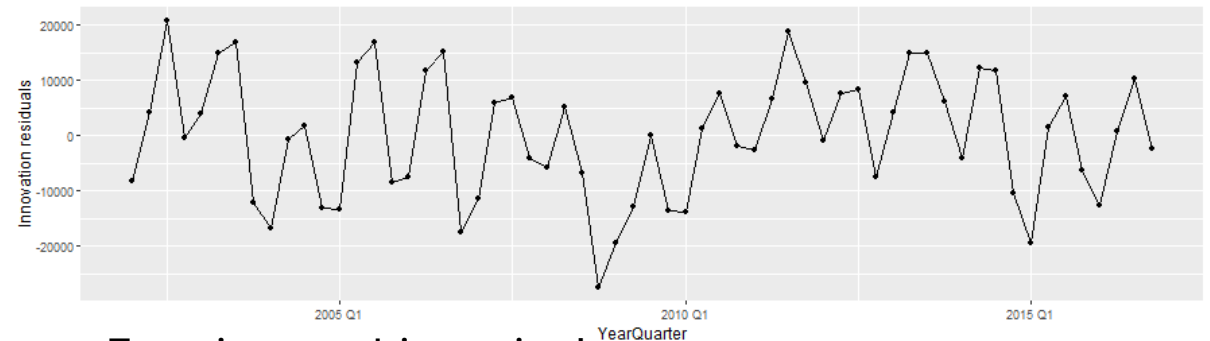
Dynamic regression: Regression for time series data

- (20) Regression with ARIMA errors (sometimes called ARIMAX)
- (30) Deterministic and stochastic trend
 - In-class exercise
- (15) Transfer function model (TFM): Dynamic regression with lagged predictors
- (15) Number of lagged predictors

Example New Car Sales

- Predictors: $x_1 = \text{DPIPC}$, $x_2 = \text{UR}$, $x_3 = \text{PR}$, $x_4 = \text{UMICS}$
- **OLS (ordinary least squares) regression** $y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \beta_3 x_{3,t} + \beta_4 x_{4,t} + \epsilon_t$
- Error assumption: $\epsilon_t \sim \text{NID}(0, \sigma_e^2)$
- From the residual analysis, clearly errors are not white noise.

```
nsc %>%  
  model(TSLM(NCS ~ DPIPC + UR + PR + UMICS)) %>%  
  gg_tsresiduals()
```



Example New Car Sales

Regression with ARIMA errors

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \beta_3 x_{3,t} + \beta_4 x_{4,t} + \eta_t$$

- **Regression error:** $\eta_t \sim \text{ARIMA}$ with innovations (shocks) ϵ_t
- **Innovations** assumption: $\epsilon_t \sim \text{NID}(0, \sigma_e^2)$
- There are two distinct error processes.

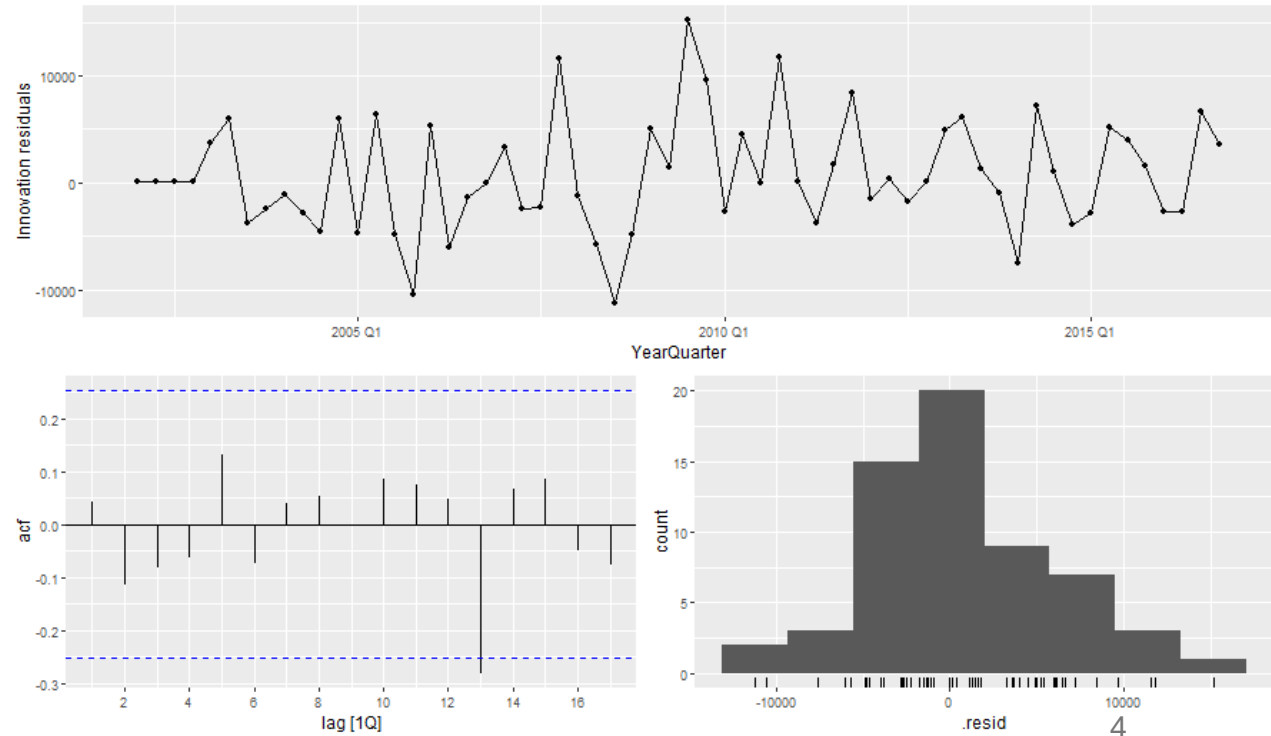
```
ncs %>%
  model(ARIMA(NCS ~ DPIPc + UR + PR + UMICS)) %>%
  gg_tsresiduals()

> ncs %>%
+   model(ARIMA(NCS ~ DPIPc + UR + PR + UMICS)) %>%
+   report()
Series: NCS
Model: LM w/ ARIMA(1,0,0)(0,1,1)[4] errors

Coefficients:
      ar1      sma1    DPIPc      UR      PR      UMICS
  0.8920  -0.4951  0.7670 -6800.838 2457.481  331.4932
```

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \beta_3 x_{3,t} + \beta_4 x_{4,t} + \eta_t$$

$$(1 - \phi_1 B)(1 - B^4)\eta_t = \epsilon_t + (1 - \theta_1 B^4)\epsilon_t$$





Regression with ARIMA errors

- Ordinary least squares (OLS) Regression model

$$y_t = \beta_0 + \beta_1 x_{1,t} + \dots + \beta_k x_{k,t} + \epsilon_t$$

- In regression, we assume that ϵ_t was white noise.
- Now, we want to allow ϵ_t to be autocorrelated.

ARIMA(p, d, q)(P, D, Q) $_m$ in general. Here, a nonseasonal ARIMA is written, simply for easy illustration.

- Regression model with ARIMA(p, d, q) errors

$$y_t = \beta_0 + \beta_1 x_{1,t} + \dots + \beta_k x_{k,t} + \eta_t$$

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon_t$$

- η_t regression errors
- ϵ_t Innovations (shocks)

where ϵ_t is white noise

	Name	Description
η_t	Regression errors	The serially correlated component of the regression error term
ϵ_t	Innovations/Shocks/White noise	Random disturbances $\sim NID$ with mean 0 and constant variance

Why dynamic regression (regression with ARIMA errors)?

- OLS regression may **exaggerate significance**.
 - OLS assumes white-noise errors. But the residuals are autocorrelated.
- After allowing ARIMA errors, only **UR** and **UMICS** remain significant.
- Compare OLS and dynamic regression models using out-of-sample forecast accuracy.

```
> ncs %>%  
+   model(regOLS = TSLM(NCS ~ DPIPC + UR + PR + UMICS),  
+         regARIMA = ARIMA(NCS ~ DPIPC + UR + PR + UMICS)) %>%  
+   tidy()  
# A tibble: 11 x 6
```

	.model	term	estimate	std.error	statistic	p.value
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	regOLS	(Intercept)	25428.	37466.	0.679	5.00e- 1
2	regOLS	DPIPC	3.23	0.392	8.24	3.52e-11 *
3	regOLS	UR	-7667.	1662.	-4.61	2.42e- 5 *
4	regOLS	PR	-3219.	1413.	-2.28	2.66e- 2 *
5	regOLS	UMICS	1122.	211.	5.33	1.91e- 6 *
6	regARIMA	ar1	0.892	0.0721	12.4	1.20e-17
7	regARIMA	sma1	-0.495	0.140	-3.53	8.29e- 4
8	regARIMA	DPIPC	0.767	1.62	0.472	6.39e- 1
9	regARIMA	UR	-6801.	2326.	-2.92	4.98e- 3 *
10	regARIMA	PR	2457.	2389.	1.03	3.08e- 1
11	regARIMA	UMICS	331.	146.	2.27	2.69e- 2 *

Stochastic and deterministic trends

Two different ways of modeling a linear trend

A deterministic trend is obtained using the regression model

$$y_t = \beta_0 + \beta_1 t + \eta_t$$

where η_t is an ARMA process.

A stochastic trend is obtained using the model

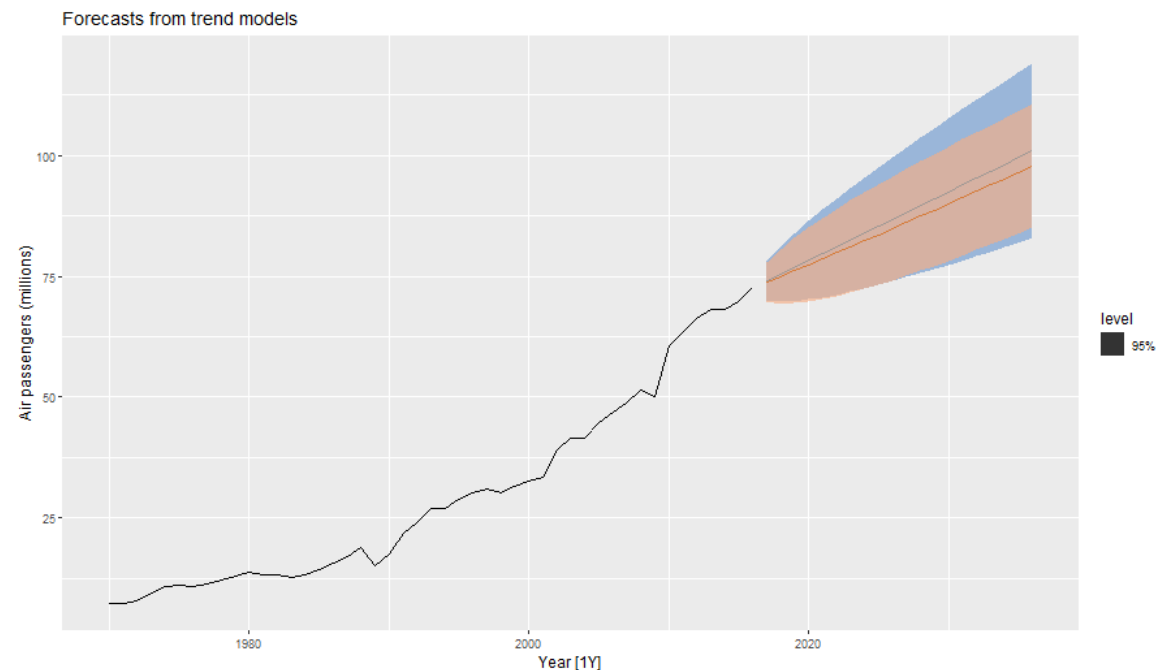
$$y_t = \beta_0 + \beta_1 t + \eta_t$$

where η_t is an ARIMA process with $d = 1$.

Stochastic and deterministic trends

- Although the growth estimates are similar, the prediction intervals are not.
- Safer to forecast with stochastic trends, especially for longer forecast horizons, as the prediction intervals allow for greater uncertainty in future growth.

```
#####  
# Stochastic and deterministic trend  
#####  
aus_airpassengers |>  
  autoplot(Passengers) +  
  labs(y = "Passengers (millions)",  
        title = "Total annual air passengers")  
  
fit_deterministic <- aus_airpassengers |>  
  model(deterministic = ARIMA(Passengers ~ 1 + trend() +  
    pdq(d = 0)))  
  
fit_stochastic <- aus_airpassengers |>  
  model(stochastic = ARIMA(Passengers ~ pdq(d = 1)))  
  
aus_airpassengers |>  
  autoplot(Passengers) +  
  autolayer(fit_stochastic |> forecast(h = 20),  
    colour = "#0072B2", level = 95) +  
  autolayer(fit_deterministic |> forecast(h = 20),  
    colour = "#D55E00", alpha = 0.65, level = 95) +  
  labs(y = "Air passengers (millions)",  
        title = "Forecasts from trend models")
```



Stochastic and deterministic trends

- OLS regression trend alone is not enough when residuals are autocorrelated.
- Deterministic trend with ARIMA errors is much better.
- Stochastic trend model gives the best test performance among the three error measures, including the lowest continuous-rank probability score (CRPS).

```
# Train-test split for out-of-sample score

train <- aus_airpassengers |>
  filter_index(. ~ "2015")

test <- aus_airpassengers |>
  filter_index("2016" ~ .)

fit <- train |>
  model(
    OLSreg = TSLM(Passengers ~ trend()),
    deterministic = ARIMA(Passengers ~ 1 + trend() + pdq(d = 0)),
    stochastic = ARIMA(Passengers ~ pdq(d = 1))
  )

fc <- fit |>
  forecast(new_data = test)

fc |>
  accuracy(
    test,
    measures = list(
      ME = ME,
      RMSE = RMSE,
      MAE = MAE,
      MAPE = MAPE,
      crps = CRPS
    )
  )
```

.model	.type	ME	RMSE	MAE	MAPE	crps
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
OLSreg	Test	11.4	11.4	11.4	15.7	8.06
deterministic	Test	1.68	1.68	1.68	2.31	0.999
stochastic	Test	1.43	1.43	1.43	1.97	0.864

R: in-class exercise

Objective:

Modify an existing forecasting workflow in R

Task:

Add three more models and compare their test-set accuracy including CRPS.

- 1) Trend regression with white-noise errors
- 2) Deterministic trend with ARMA error
- 3) Stochastic trend with ARIMA error with $d = 1$

Set	t	Date	Demand	FC.Mean	FC.Naive	FC.Seasonal Naïve	FC.Drift
Train	1	2001-01-01	190				
Train	2	2001-04-01	260				
Train	3	2001-07-01	330				
Train	4	2001-10-01	480				
Train	5	2002-01-01	570				
Train	6	2002-04-01	710				
Train	7	2002-07-01	650				
Train	T=8	2002-10-01	690				
Test	9	2003-01-01	730	485	690	570	761
Test	10	2003-04-01	680	485	690	710	833
Test	11	2003-07-01	660	485	690	650	904
Test	12	2003-10-01	590	485	690	690	976
Test MAE				180	45	75	204

```
# in-class exercise
```

```
myts <- tsibble(
  Quarter = yearquarter("2001 Q1") + 0:11,
  Demand = c(190,260,330,480,570,710,650,690,730,680,660,590)
  index = Quarter
)

myts_train <- myts %>%
  filter(year(Quarter) <= 2002)

myts_fit <- myts_train %>%
  model(
    Mean = MEAN(Demand),
    Naive = NAIVE(Demand),
    "Seasonal Naive" = SNAIVE(Demand),
    Drift = NAIVE(Demand ~ drift())
  )

myts_fc <- myts_fit %>%
  forecast(h=4)

myts_fc %>%
  accuracy(
    myts,
    measures = list(
      ME = ME, RMSE = RMSE, MAE = MAE, MAPE = MAPE
    )
  )
```

.model	.type	ME	RMSE	MAE	MAPE
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Drift	Test	-204.	241.	204.	32.3
Mean	Test	180	187.	180	26.6
Naive	Test	-25	56.1	45	7.11
Seasonal Naive	Test	10	95.7	75	11.2

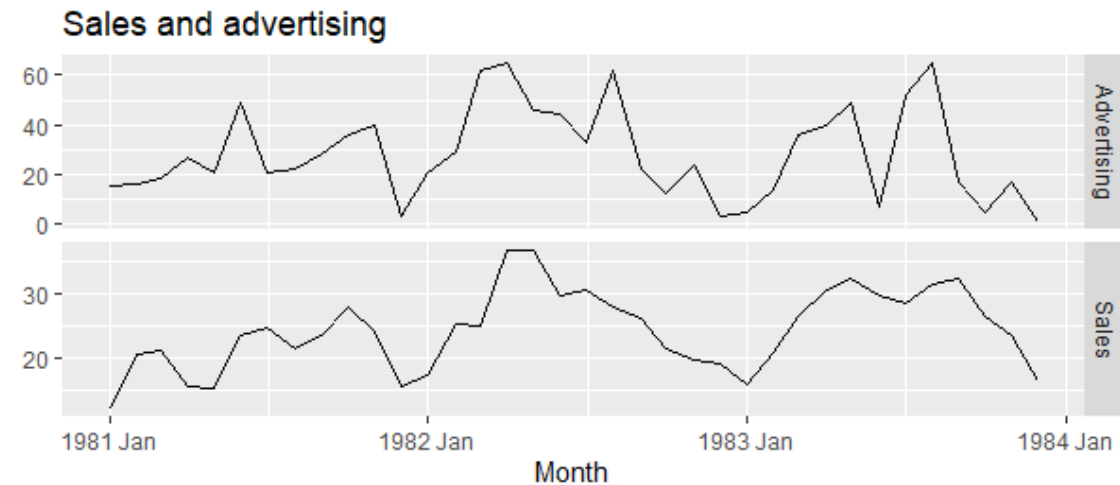
Transfer function models (TFM)

Box-Jenkins (1970s) terminology

Lagged predictors

Example: Blattberg and Jeuland (1981)

- Sales volume (y_t) and advertising expenditure (x_t) for a dietary weight control product.
- Sales are measured in equivalent serving units.
- Advertising is measured in dollars.
- Sales y_t may be a function of
 - current advertising x_t and
 - advertising in past few months, $x_{t-1}, x_{t-2}, \dots$



TFM

Lagged predictors

- Recall in MA model, we have memory from innovations $\epsilon_{t-1}, \epsilon_{t-2}, \dots$
- Sometimes a change in x_t does not affect y_t instantaneously
- Dynamic regression model, which allows for lagged effects, can be written as

$$y_t = \beta_0 + v_0 x_t + v_1 x_{t-1} + v_2 x_{t-2} + \dots + v_k x_{t-k} + \eta_t$$

where η_t is an ARIMA process

- Examples

y_t	x_t	Why Lag Matters
SKU sales	Price/promotion	Promotion lifts sales now; stockpiling may reduce future sales
Supplier lead time	Purchase orders released	High order volume can strain supplier capacity and lengthen lead time later
Output rate	Machine maintenance schedule	Preventive maintenance affects production capacity with delay
Port congestion	Export volumes	Why?
Return volume	Sales volume	Why?



Example: Blattberg and Jeuland (1981)

$$y_t = \beta_0 + v_0 x_t + v_1 x_{t-1} + v_2 x_{t-2} + v_3 x_{t-3} + v_4 x_{t-4} + v_5 x_{t-5} + \eta_t$$

$$y_t = 17.9551 + 0.1133x_t + 0.1583x_{t-1} + 0.0344x_{t-2} + 0.0218x_{t-3} - 0.014x_{t-4} - 0.0714x_{t-5} + \eta_t$$

where η_t is ARIMA(0,0,1) error process

```
fit <- dt %>%  
  model(ARIMA(Sales ~ pdq(d=0) + Advertising +  
              lag(Advertising) + lag(Advertising,2) +  
              lag(Advertising,3) + lag(Advertising,4) + lag(Advertising,5)))  
report(fit)
```

- Note how the effect of advertising on sales is positive and drops off over time: $v_1 > v_2 > v_3 > 0$
- The values of $v_0, v_1, \dots$ are called the **impulse response weights** (or **transfer function weights**).
- The coefficient v_i is a measure of how y_t responds to a change in x_{t-i}
- After some time, the effect of advertising will become so small that it will be swamped by the random error. Here, v_4, v_5 negative by chance.

Transfer function models (TFM)

- Suppose that we have only one predictor in our model.
- Then a model which allows for lagged effects can be written as

$$y_t = \beta_0 + v_0 x_t + v_1 x_{t-1} + v_2 x_{t-2} + \cdots + v_k x_{t-k} + \eta_t$$

where η_t is an ARIMA process

- Rewrite the model as

$$\begin{aligned} y_t &= \beta_0 + (v_0 + v_1 B + v_2 B^2 + \cdots + v_k B^k) x_t + \eta_t \\ &= \beta_0 + v(B) x_t + \eta_t \end{aligned}$$

- $v(B)$ is called a **transfer function** since it describes how change in x_t is transferred to y_t
 - A generalization of dynamic regression models, known as “transfer function model” (Box & Jenkins 1970).
 - In a full **Transfer Function Model**, we model y_t as a potentially a function of past lags of y_t and current and past lags of the x -variables.

Number of lagged predictors

Lagged predictors

- Consider the model

$$\begin{aligned}y_t &= \beta_0 + (\nu_0 + \nu_1 B + \nu_2 B^2 + \cdots + \nu_k B^k)x_t + \eta_t \\ &= \beta_0 + \nu(B)x_t + \eta_t\end{aligned}$$

where η_t is an ARMA process

- The values of k can be selected using the AICc, along with the values of p and q for the ARMA error.

Transfer function model

$$y_t = \beta_0 + (v_0 + v_1 B + v_2 B^2 + \dots + v_k B^k) x_t + \eta_t$$

Example: TV advertising and insurance quotations

Consider including advertising expenditure for up to **4 months**; that is, the model may include advertising expenditure in the current month, and the **three** months before that.

Model 0:

$$y_t = \beta_0 + \underset{\text{Current month}}{v_0 x_t} + \eta_t$$

Model 1:

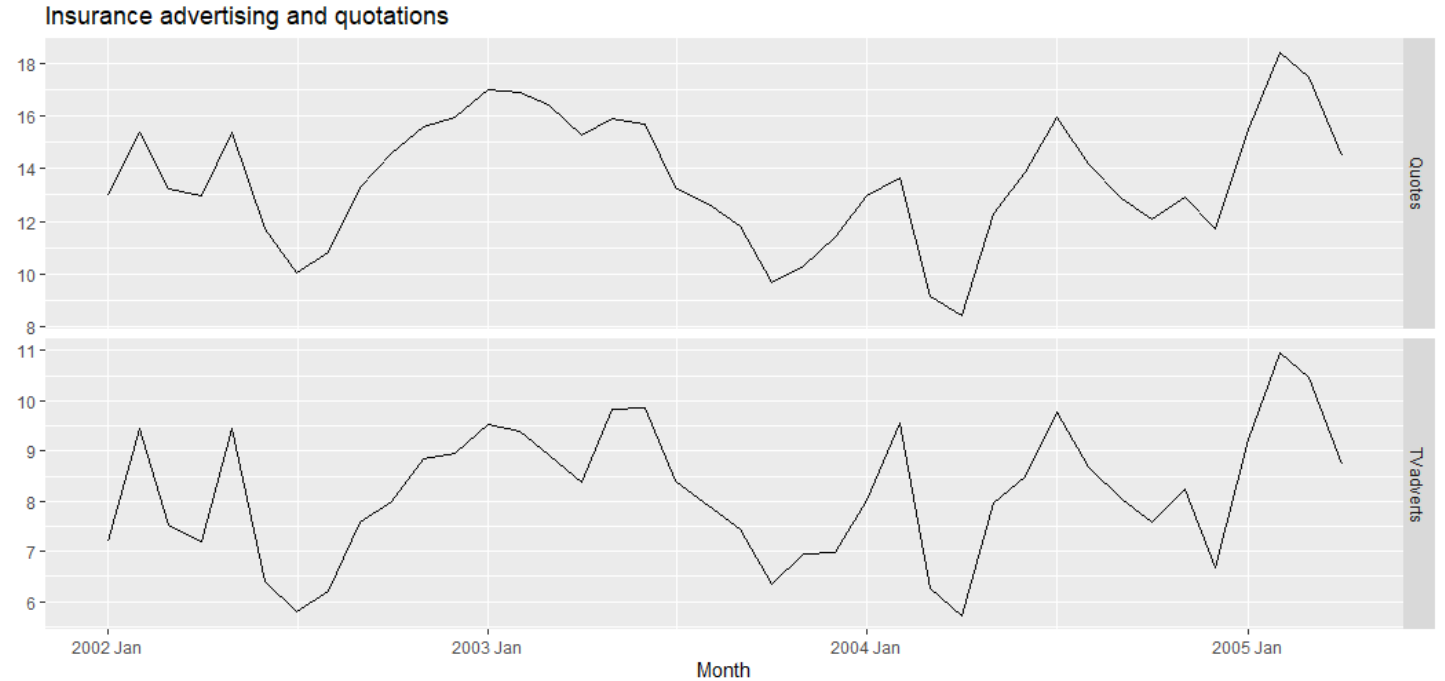
$$y_t = \beta_0 + (v_0 + v_1 B) x_t + \eta_t$$

$$y_t = \beta_0 + \underset{\text{Current month, 1 month before than}}{v_0 x_t + v_1 x_{t-1}} + \eta_t$$

Model 2:

Model 3:

$$y_t = \beta_0 + (v_0 + v_1 B + v_2 B^2 + v_3 B^3) x_t + \eta_t$$



```
> insurance
# A tsibble: 40 x 3 [1M]
  Month Quotes TVadverts
  <mtch> <dbl> <dbl>
1 2002 Jan 13.0 7.21
2 2002 Feb 15.4 9.44
3 2002 Mar 13.2 7.53
4 2002 Apr 13.0 7.21
5 2002 May 15.4 9.44
6 2002 Jun 11.7 6.42
7 2002 Jul 10.1 5.81
8 2002 Aug 10.8 6.20
9 2002 Sep 13.3 7.59
10 2002 Oct 14.6 8.00
# ... with 30 more rows
```

```
insurance %>%
  pivot_longer(Quotes:TVadverts) %>%
  ggplot(aes(x = Month, y = value)) +
  geom_line() +
  facet_grid(vars(name), scales = "free_y") +
  labs(y = "", title = "Insurance advertising and quotations")
```

Transfer function model $y_t = \beta_0 + (v_0 + v_1B + v_2B^2 + \dots + v_kB^k)x_t + \eta_t$

The values of k can be selected using the AICc

Example: TV advertising and insurance quotations

Recall our four models

- Model 0: $y_t = \beta_0 + v_0x_t + \eta_t$
- Model 1: $y_t = \beta_0 + (v_0 + v_1B)x_t + \eta_t$
- ...
- Model 3: $y_t = \beta_0 + (v_0 + v_1B + v_2B^2 + v_3B^3)x_t + \eta_t$

Model 3 training set starts from 2002 Apr, i.e., the first fitted value is Quotes 2002 Apr

```
> insurance
# A tsibble: 40 x 3 [1M]
  Month Quotes TVadverts
  <mt> <dbl> <dbl>
1 2002 Jan 13.0 7.21
2 2002 Feb 15.4 9.44
3 2002 Mar 13.2 7.53
4 2002 Apr 13.0 7.21
5 2002 May 15.4 9.44
6 2002 Jun 11.7 6.42
7 2002 Jul 10.1 5.81
8 2002 Aug 10.8 6.20
9 2002 Sep 13.3 7.59
10 2002 Oct 14.6 8.00
# ... with 30 more rows
```

Unfair comparisons:
Model 0 ... Model 3
different training sets

Model 0 first fitted
value is _____

Model 3 first fitted
value is _____

When comparing models, it is important that they all use the same training set.

Exclude the first three months in order to make fair comparisons.

```
insurance %>%
  # Restrict data so models use same fitting period
  mutate(Quotes = c(NA, NA, NA, Quotes[4:40]))
```

```
> insurance %>%
+ # Restrict data so models use same fitting period
+ mutate(Quotes = c(NA, NA, NA, Quotes[4:40]))
# A tsibble: 40 x 3 [1M]
  Month Quotes TVadverts
  <mt> <dbl> <dbl>
1 2002 Jan NA 7.21
2 2002 Feb NA 9.44
3 2002 Mar NA 7.53
4 2002 Apr 13.0 7.21
5 2002 May 15.4 9.44
6 2002 Jun 11.7 6.42
7 2002 Jul 10.1 5.81
8 2002 Aug 10.8 6.20
9 2002 Sep 13.3 7.59
10 2002 Oct 14.6 8.00
# ... with 30 more rows
```

Fair comparisons:
Model 0 ... Model 3
same training sets

Example: TV advertising and insurance quotations

```
fit <- insurance %>%
  # Restrict data so models use same fitting period
  mutate(Quotes = c(NA, NA, NA, Quotes[4:40])) %>%

  # Estimate models
  model(
    lag0 = ARIMA(Quotes ~ pdq(d = 0) + TVadverts),
    lag1 = ARIMA(Quotes ~ pdq(d = 0) +
                  TVadverts + lag(TVadverts)),
    lag2 = ARIMA(Quotes ~ pdq(d = 0) +
                  TVadverts + lag(TVadverts) +
                  lag(TVadverts, 2)),
    lag3 = ARIMA(Quotes ~ pdq(d = 0) +
                  TVadverts + lag(TVadverts) +
                  lag(TVadverts, 2) + lag(TVadverts, 3))
  )

glance(fit)
```

```
> glance(fit)
# A tibble: 4 x 8
  .model sigma2 log_lik   AIC   AICc   BIC ar_roots ma_roots
  <chr>   <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
1 lag0    0.265   -28.3  66.6  68.3  75.0 <cp1 [2]> <cp1 [0]>
2 lag1    0.209   -24.0  58.1  59.9  66.5 <cp1 [1]> <cp1 [1]>
3 lag2    0.215   -24.0  60.0  62.6  70.2 <cp1 [1]> <cp1 [1]>
4 lag3    0.206   -22.2  60.3  65.0  73.8 <cp1 [1]> <cp1 [1]>
```

- Choose the optimal lag length for advertising based on the AICc.
- The best model (with the smallest AICc value) is lag1 with two predictors; that is, it includes advertising only in the current month and the previous month.

Example: TV advertising and insurance quotations

```
fit_best <- insurance %>%  
  model(ARIMA(Quotes ~ pdq(d = 0) +  
    TVadverts + lag(TVadverts)))  
  
report(fit_best)
```

```
> report(fit_best)  
Series: Quotes  
Model: LM w/ ARIMA(1,0,2) errors  
  
Coefficients:  
      ar1      ma1      ma2 TVadverts lag(TVadverts) intercept  
0.5123  0.9169  0.4591    1.2527         0.1464      2.1554  
s.e.  0.1849  0.2051  0.1895    0.0588         0.0531      0.8595  
  
sigma^2 estimated as 0.2166: log likelihood=-23.94  
AIC=61.88 AICc=65.38 BIC=73.7
```

Now re-estimate that model, but using all the available data.

Let

y_t = the number of quotations provided in month t

x_t = the advertising expenditure in month t

The model can be written as

$$y_t = 2.155 + 1.253x_t + 0.146x_{t-1} + \eta_t$$

$$\eta_t = 0.512\eta_{t-1} + \epsilon_t + 0.917\epsilon_{t-1} + 0.459\epsilon_{t-2}$$

$$\epsilon_t \sim NID(0, \hat{\sigma}^2 = 0.2166)$$

- backshift notation $(1 - \phi_1 B)\eta_t = (1 + \theta_1 B + \theta_2 B^2)\epsilon_t$
- Wold decomposition: ratio of two finite lag polynomials

$$\eta_t = \frac{\phi_1(B)}{\theta_2(B)} \epsilon_t$$

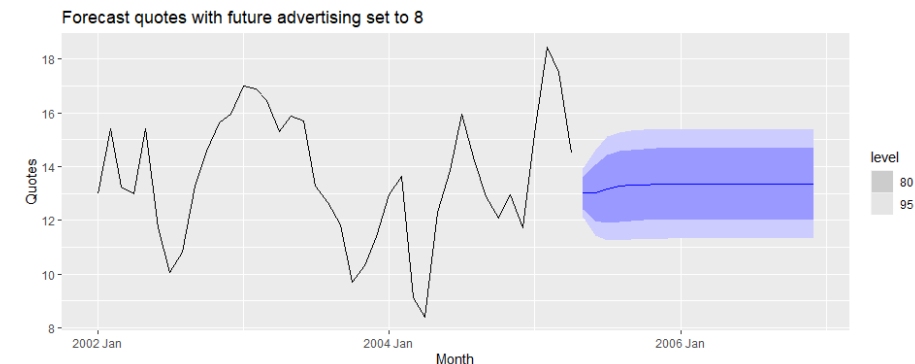
Example: TV advertising & insurance quotations

- Forecasting with dynamic regression requires future values of predictors.
 - Some are known in advance, e.g., holidays, time trend, quarter.
 - Some must be forecast first, e.g., economic indicators or market demand.
 - Some are internal planning decisions, e.g., promotion, price, advertising, or capacity plans.
- Sales and Operations Planning (S&OP)** provides the assumptions needed for future predictors, such as advertising, promotion, price, or capacity plans.

```
insurance_future <- new_data(insurance, 20) %>%
  mutate(TVadverts = 8)      If we set the 20 future monthly advertising to 8 units,
                              we get the following forecasts.

fit_best %>%
  forecast(insurance_future) %>%
  autoplot(insurance) +
  labs(
    y = "Quotes",
    title = "Forecast quotes with future advertising set to 8"
  )
```

```
> insurance_future
# A tsibble: 20 x 2 [1M]
      Month TVadverts
  <mtch>    <dbl>
1 2005 May      8
2 2005 Jun      8
3 2005 Jul      8
4 2005 Aug      8
5 2005 Sep      8
6 2005 Oct      8
7 2005 Nov      8
8 2005 Dec      8
9 2006 Jan      8
10 2006 Feb     8
11 2006 Mar     8
12 2006 Apr     8
13 2006 May     8
14 2006 Jun     8
15 2006 Jul     8
16 2006 Aug     8
17 2006 Sep     8
18 2006 Oct     8
19 2006 Nov     8
20 2006 Dec     8
```



Transfer function model

Example New Car Sales

- Transfer function models describe dynamic effects of predictors.
- A predictor may affect the response:
 - immediately,
 - after a delay,
 - or gradually over time.
- With multiple predictors, each predictor can have a different dynamic pattern.
- Lag structure should be guided by business logic first, then checked by forecast performance and diagnostics.

```
> ncs %>%
+   model(
+     mTFM_NCS = ARIMA(NCS ~ pdq(0) + UR + lag(UR) + UMICS + lag(UMICS) + lag(UMICS, 2))
+   ) %>%
+   tidy()
# A tibble: 8 x 6
  .model term estimate std.error statistic p.value
  <chr> <chr> <dbl> <dbl> <dbl> <dbl>
1 mTFM_NCS ma1 0.810 0.0841 9.62 1.26e-13
2 mTFM_NCS sar1 0.912 0.0457 20.0 1.10e-27
3 mTFM_NCS UR -11237. 2888. -3.89 2.60e- 4
4 mTFM_NCS lag(UR) 4603. 2719. 1.69 9.58e- 2
5 mTFM_NCS UMICS 375. 130. 2.89 5.46e- 3
6 mTFM_NCS lag(UMICS) 221. 131. 1.68 9.84e- 2
7 mTFM_NCS lag(UMICS, 2) 269. 137. 1.97 5.39e- 2
8 mTFM_NCS intercept 146340. 22901. 6.39 3.07e- 8
```

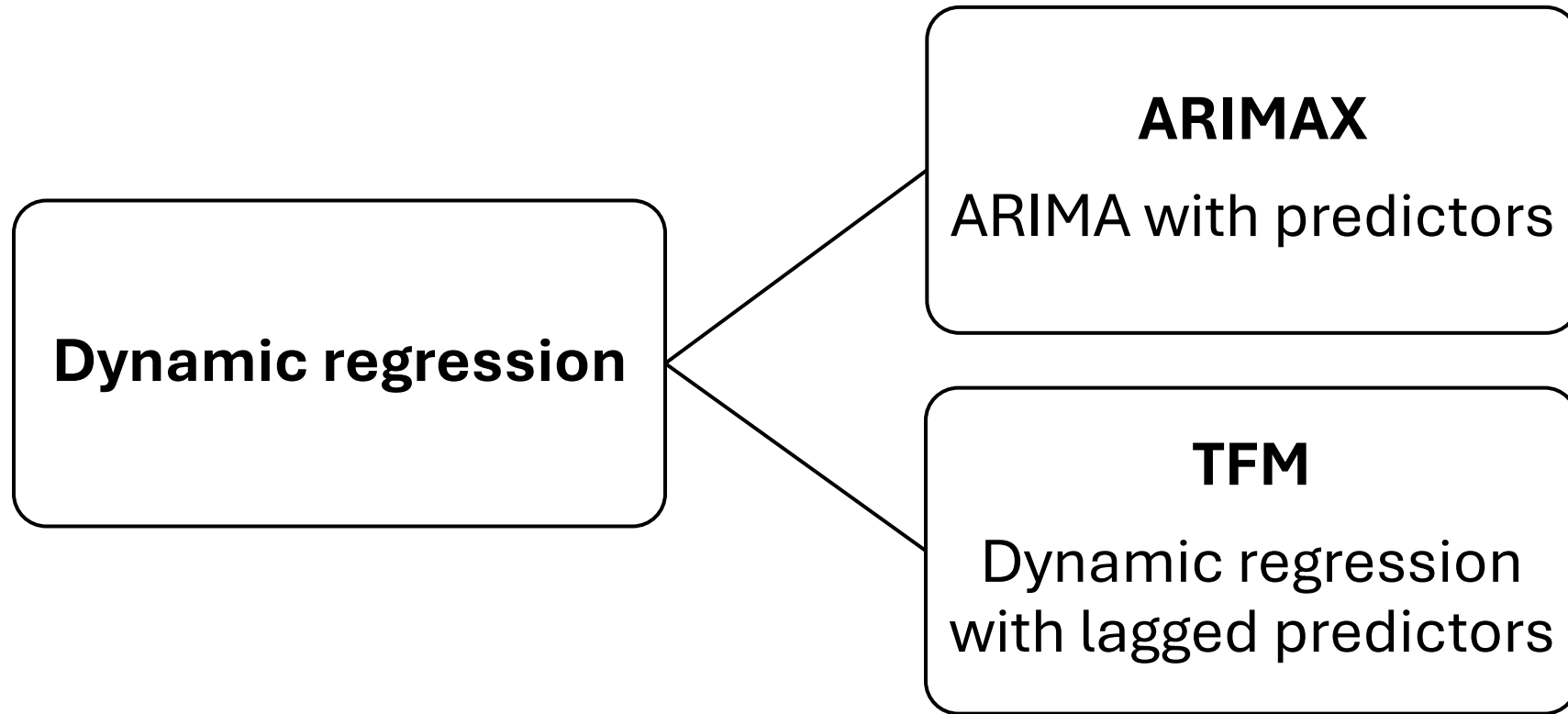
$$NCS_t = \text{baseline} + UR_t + UR_{t-1} + UMICS_t + UMICS_{t-1} + UMICS(t-2) + \eta_t$$

Where η_t follows $ARIMA(0,0,1)(1,0,0)_4$

Possible dynamic story

- UR: labor-market conditions may affect car purchases with delay.
- UMICS: consumer sentiment may lead actual purchase decisions..

In this class,...



Terminology note: ARIMAX, dynamic regression, and TFM (reference only)*

	Dynamic regression connects regression and ARIMA ideas				TFM
	ARMA	Regression	Regression with ARMA errors	ARIMAX	
	$y_t = \frac{\theta(B)}{\phi(B)} \epsilon_t$	$y_t = \beta x_t + \epsilon_t$	$y_t = \beta x_t + \frac{\theta(B)}{\phi(B)} \epsilon_t$	$y_t = \frac{\beta}{\phi(B)} x_t + \frac{\theta(B)}{\phi(B)} \epsilon_t$	$y_t = \frac{\omega(B)}{\delta(B)} x_t + \frac{\theta(B)}{\phi(B)} \epsilon_t$
Error ϵ_t	White noise (uncorrelated)		ARMA-type error (autocorrelated)		
Description			Regression + ARMA errors	Adds autoregressive terms of y_t ($y_{t-1}, y_{t-2}, \dots$)	Add lagged predictors $x_{t-1}, x_{t-2}, \dots$
	No predictor, only past shocks explain y_t	x_t explains y_t , errors are uncorrelated	x_t explains y_t but errors are autocorrelated	y_t depends on contemporaneous variable (current predictor) x_t and its own past values $y_{t-\ell}$	Most general: y_t depends on current x_t and distributed lag effects of predictors ($x_{t-1}, x_{t-2}, \dots$) plus $y_{t-\ell}$ (past values of y_t)
Example	Sales depend on past sales and shocks	Sales depend on current predictors + white-noise errors	Sales depend on current predictors + ARIMA errors	Monthly advertising & Sales: Sales y_t depend on this month's advertising x_t and past sales $y_{t-\ell}$	Advertising with lags: Sales y_t depend on advertising this month x_t and past 3 months ($x_{t-1}, x_{t-2}, x_{t-3}$) plus past sales $y_{t-\ell}$

Different books and R packages use these terms differently. The `arimax()` function from the TSA package in R fits the transfer function model. For this class, use dynamic regression as the main term.